

# AP CSP | Lesson 08: Data Analysis & Evidence

## Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

### Questions 1-6 | 1 mark each

#### Q1. Answer: B

B follows directly from the increased sample count. A confuses time sampling with value levels. C asserts an unsupported guarantee for an arbitrary signal. D confuses total sample count with per-sample code length; four levels can still use two bits each.

#### Q2. Answer: D

D permits exact recovery of the original digital data. A may discard needed information. B changes the dataset and is not justified by compression. C imposes an impossible universal size guarantee; some inputs cannot be made smaller by a given lossless method.

#### Q3. Answer: A

A distinguishes an unusual value from an invalid one and preserves the analysis trail. B chooses data to obtain a desired statistic. C changes meaning without evidence or documentation. D turns a single response into an unsupported population claim.

#### Q4. Answer: C

C is correct because sorting changes order, not membership or values. A confuses reordering with changing data. B describes filtering or selecting a subset. D is an exclusion, not a consequence of sorting.

#### Q5. Answer: B and D

B directly compares the three group means. D follows from  $4 + 3 + 6 = 13$ . A makes an individual causal claim not established by observational group summaries. C contradicts the different group counts. Award one mark only for B and D together with no other selections.

#### Q6. Answer: C

C is correct:  $(8 + 10 + 12 + 15) / 4 = 45 / 4 = 11.25$ . A is one observation. B is the median, the average of the middle two sorted values. D is the sum rather than the mean.

### Written-response answers | Questions 7-12

#### Q7. 4 marks

**Answer:** (a)  $40 \times 3 = 120$  bits = 15 bytes. (b)  $80 \times 3 = 240$  bits. (c) The recording still uses finitely many measurement times and finite value precision, so information between samples or finer value detail may be omitted.

**1 mark:** Calculates 120 payload bits.

**1 mark:** Converts 120 bits to 15 bytes.

**1 mark:** Calculates 240 bits after the sample count doubles.

**1 mark:** Rejects guaranteed exactness using finite sampling or finite value precision.

**Q8. 4 marks**

**Answer:** (a) Lossless saves 300 bytes; lossy saves 900 bytes. (b) The lossless option meets exact recovery, because decompression reproduces the original digital data. (c) A preview image may use lossy compression if the loss of detail is acceptable for the viewing purpose.

**1 mark:** Calculates both savings correctly: 300 and 900 bytes.

**1 mark:** Selects the lossless file for exact recovery.

**1 mark:** Explains that lossless decompression preserves the original digital records exactly.

**1 mark:** Gives a plausible lossy use and explicitly conditions it on acceptable information loss.

**Q9. 4 marks**

**Answer:** (a) Standardize “ bus ” to “Bus”. (b) Thirteen records: remove one duplicate and exclude the missing and negative times. (c) Unusual does not mean erroneous; no error is confirmed. (d) Record an exclusion with its ID and reason, or another reproducible rule or transformation.

**1 mark:** Standardizes the label to Bus without changing its meaning.

**1 mark:** Gives 13 records and accounts for the three excluded rows.

**1 mark:** Explains that 120 is unusual but has no evidence of error.

**1 mark:** Names a useful log item such as an excluded ID plus reason or a recorded standardization rule.

**Q10. 4 marks**

**Answer:** (a)  $381 / 13 \approx 29.31$  minutes. (b) Count 12; sum 261; mean  $261 / 12 = 21.75$  minutes. (c) No. The change shows sensitivity to that observation, not evidence that its value is invalid.

**1 mark:** Calculates the baseline mean as 29.31 minutes or the exact equivalent  $381/13$ .

**1 mark:** Gives comparison count 12 and sum 261 minutes.

**1 mark:** Calculates comparison mean 21.75 minutes.

**1 mark:** Rejects the error claim and distinguishes sensitivity from evidence of invalidity.

**Q11. 4 marks**

**Answer:** (a) A bar chart with mode categories Walk, Bike and Bus, and mean one-way commute time in minutes on the numeric axis. (b) For example: synthetic retained sample,  $n=13$ , with 120 retained. (c) Each bar aggregates a group into its mean, so individual values and their variation are not shown.

**1 mark:** Chooses a suitable comparison chart and identifies the three mode categories.

**1 mark:** Labels the numeric values as mean commute time in minutes.

**1 mark:** Gives a meaningful view caption detail, such as retained  $n=13$ , synthetic source or 120 retained.

**1 mark:** Explains that group means aggregate observations and do not display all individual times or spread.

**Q12. 4 marks**

**Answer:** (a) In this retained sample, Bus has the highest mean reported one-way commute time. (b) Its mean is 47.17 minutes across 6 retained responses. (c) Distance or other group differences may explain part of the association, so mode is not isolated as a cause; a single voluntary school-morning sample does not represent all students.

**1 mark:** Rewrites the claim as a descriptive comparison restricted to the retained sample.

**1 mark:** Gives an appropriate baseline statistic with units and count, such as Bus 47.17 minutes,  $n=6$ .

**1 mark:** Explains a causal limitation such as differing distance or uncontrolled conditions.

**1 mark:** Explains a population limitation such as voluntary participation, one school or one morning.